

Musterlösung Serie 8

DIMENSION AND BASIS

1. Dimension. Compute the dimension and find a basis of the following spaces

- (a) The space of upper triangular matrices in $M_{n \times n}(\mathbb{R})$ over $\mathbb{R}$ (for a definition, see Serie 7, exercise 2.b));
- (b) The space of diagonal matrices in $M_{n \times n}(\mathbb{R})$ over $\mathbb{R}$, where diagonal matrices are all matrices $A = (a_{ij})$ such that $a_{ij} = 0$ whenever $i \neq j$;
- (c) The space of symmetric matrices

$$W = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\},$$

where $\cdot^T$ denotes the operation $A = (a_{ij})_{1 \leq i, j \leq n} \mapsto (a_{ji})_{1 \leq i, j \leq n}$;

- (d) The space of matrices $A \in M_{n \times n}(\mathbb{F}_2)$ such that the sum of the columns of A is the null vector.

Lösung:

- (a) A matrix is upper triangular if its entries $a_{ij} = 0$ whenever $i > j$. For all $1 \leq i, j \leq n$ such that $i \leq j$, let A_{kl} be the matrix whose entries are zero except for the entry a_{kl} , which is set to 1. Note that the set

$$\{A_{kl} \mid 1 \leq k, l \leq n \wedge k \leq l\}$$

is linearly independent since the matrices do not share any entries and it generates the space of upper triangular matrices. Hence it is a basis. Moreover the cardinality of this set is the sum of natural numbers smaller or equal to n

$$\sum_{m=1}^n m = \frac{n(n+1)}{2}.$$

- (b) For $1 \leq k \leq n$, let A_k be the matrix whose entries are all zero, except for the entry a_{kk} , which is set to 1. The set $\{A_k \mid 1 \leq k \leq n\}$ is linearly independent and generates the space of diagonal matrices. Hence it is a basis. The cardinality of this set is n .

- (c) For $1 \leq k \leq l \leq n$, let A_{kl} be the matrix such that $a_{kl} = 1 = a_{lk}$ and all other entries are set to 0. The set $\{A_{kl} \mid 1 \leq k \leq l \leq n\}$ is linearly independent and generates the space of symmetric matrices. We have n matrices with a unique 1 on the diagonal and

$$\sum_{m=1}^{n-1} m = \frac{(n-1)n}{2},$$

additional matrices. Hence the dimension is $n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.

- (d) Let W denote the said space. For any $A = (a_{ij})_{1 \leq i, j \leq n} \in W$, we have that for any $1 \leq i \leq n$,

$$\sum_{k=1}^n a_{ik} = 0 \Leftrightarrow a_{in} = -\sum_{k=1}^{n-1} a_{ik}.$$

This shows that we have $n-1$ degrees of freedom for each row. Therefore, the dimension of W is $\leq n(n-1)$. We attempt to find a basis. For $1 \leq i \leq n$ and $1 \leq k \leq n-1$, we define $A_k^{(i)}$ to be the matrix that has a 1 in place of the (i, k) -th and $(i, k+1)$ -st entries, and whose other entries vanish. We denote S_i the set $\{A_k^{(i)} \mid 1 \leq k \leq n-1\}$. To help you visualise it, we write out S_1 :

$$S_1 = \left\{ \begin{pmatrix} 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & 0 & 0 & \cdots & 1 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \right\}.$$

We first show that any S_i is linearly independent. Consider a vanishing linear combination in S_i . Note that the coefficients in front of $A_1^{(i)}$ and $A_{n-1}^{(i)}$ must vanish since $A_1^{(i)}$ is the only matrix with an entry in the first column and $A_{n-1}^{(i)}$ is the only matrix with an entry in the last column. Repeat the argument in $T \setminus \{A_1^{(i)}, A_{n-1}^{(i)}\}$ for $A_2^{(i)}$ and $A_{n-2}^{(i)}$, and so on. This proves that the only vanishing linear combination in S_i is the trivial one.

Now, let $S = \bigcup_{i=1}^n S_i$. We show that S is linearly independent. Assume that we have a vanishing linear combination in S . Since the i -th row of the linear combination vanishes, the combination of elements of S_i vanishes in particular. By the previous paragraph, this implies that the elements of S_i have vanishing coefficients. Since this holds for all $1 \leq i \leq n$, the only vanishing linear combination in S is the trivial one. Hence S is linearly independent.

Since $\dim(W) \leq n(n-1)$ and S is a linearly independent subset of this size, it is a basis.

2. Basis of a linear subspace. Let W be the linear subspace generated by the vectors

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 4 \\ 2 \\ 3 \end{pmatrix}, v_4 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 1 \end{pmatrix}.$$

- (a) Compute the dimension of W .
- (b) Determine a system of equations whose solution is W .

Hint. If the dimension of W is 3 and you find a vector n that is orthogonal to all $w \in W$, i.e. such that

$$0 = \left\langle \begin{pmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix} \right\rangle := \sum_{i=1}^4 n_i w_i$$

for all $w \in W$, then W is the solution to the system

$$n_1 x_1 + n_2 x_2 + n_3 x_3 + n_4 x_4 = 0.$$

- (c) Find all the possible bases of W that can be built using v_1, v_2, v_3, v_4 . How many are there?

Solution:

- (a) Gaussian Elimination yields that v_1, v_2 and v_4 are linearly independent. The vector $v_3 = v_1 + v_4$ is a linear combination of v_1 and v_4 . Hence W is a three-dimensional hyperplane in $\mathbb{R}^4$.
- (b) It is thus enough to determine the normal vector $n := (n_1, n_2, n_3, n_4) \in \mathbb{R}^4$ which is orthogonal on v_1, v_2, v_4 . For this, we solve the system of equations

$$\begin{cases} \langle n, v_1 \rangle = 0 \\ \langle n, v_2 \rangle = 0 \\ \langle n, v_4 \rangle = 0 \end{cases}$$

where $\langle n, v_i \rangle$ denotes the standard scalar product in $\mathbb{R}^4$. This system of equations is equivalent to

$$\begin{cases} n_1 + 2n_2 + n_3 + 2n_4 = 0 \\ \quad \quad \quad -n_3 + n_4 = 0 \\ \quad \quad \quad 2n_2 + n_3 + n_4 = 0 \end{cases}$$

Solving yields that the solutions are given by

$$n = (n_1, n_1, -n_1, -n_1)$$

for an arbitrary $n_1 \in \mathbb{R}$. As the length of a normal vector is irrelevant, we choose $n_1 = 1$ and get $n = (1, 1, -1, -1)$. Now a system of equations whose set of solutions is W is given by

$$\langle n, x \rangle = 0,$$

where $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. This is because all vectors

$$x = (x_1, x_2, x_3, x_4)$$

which satisfy this condition (and thus are solutions of this system of equations), are orthogonal to n and so are contained in W . Explicitly, we get the equation

$$x_1 + x_2 - x_3 - x_4 = 0.$$

- (c) Because of the relation $v_3 = v_1 + v_4$, the vectors v_1, v_3 and v_4 are linearly dependent, but every two out of the three are linearly independent. The vector v_2 is linearly independent from the three others. Thus v_2 must be contained in any basis which is only allowed to contain the vectors v_1, v_2, v_3, v_4 . Thus there are three possibilities:

$$\mathcal{B}_1 = \{v_1, v_2, v_3\}, \quad \mathcal{B}_2 = \{v_1, v_2, v_4\}, \quad \mathcal{B}_3 = \{v_2, v_3, v_4\}.$$

3. Dimension. Determine a basis and the dimension of the following spaces:

- (a) The set of solutions $S \subseteq \mathbb{R}^3$ of

$$\begin{aligned} x + y - z &= 0 \\ 3x + y + 2z &= 0 \\ 2x + 3z &= 0 \end{aligned}$$

- (b) $\{0\}$;
 (c) $\{(z, w) \in \mathbb{C}^2 \mid z + iw = 0\}$ as a vector space over $\mathbb{C}$;
 (d) $\{(z, w) \in \mathbb{C}^2 \mid z + iw = 0\}$ as a vector space over $\mathbb{R}$.

Lösung:

- (a) Gaussian elimination yields as the set of solutions the one-dimensional vector space $V := \left\{ \left(-\frac{3}{2}z, \frac{5}{2}z, z \right) \mid z \in \mathbb{R} \right\}$. A basis is for example formed by the vector $(-3, 5, 2)$.
 (b) We can write the zero-vector as empty sum, that is, the sum over no elements. Thus the empty set $\emptyset$ forms a basis of $\{0\}$ and the vector space has dimension 0. (Note: The zero-vector 0 can not be contained in any basis, as it is not linearly independent.)

- (c) Let V be the vector space $\{(x, y) \in \mathbb{C}^2 \mid x + iy = 0\}$ over $\mathbb{C}$. Let $(x, y) \in V$. Then we have $x + iy = 0 \Rightarrow y = ix$ and thus $(x, y) = (x, ix) = x(1, i)$ for every $(x, y) \in V$. Hence, we can write each $(x, y) \in V$ as multiple of the vector $(1, i)$. This vector is of course linearly independent in $\mathbb{C}^2$ and thus forms a basis of V . The complex dimension of this vectorspace therefore is 1.
- (d) We have shown in d) that every element in V is of the form $x(1, i)$, $x \in \mathbb{C}$. When we now consider $x = a + ib$, $a, b \in \mathbb{R}$, we get

$$x(1, i) = (a + ib)(1, i) = a(1, i) + ib(1, i) = a(1, i) + b(i, -1).$$

Hence every $(x, y) \in V$ can be written as linear combination of $(1, i)$ and $(i, -1)$. Those two are linearly independent in V over $\mathbb{R}$ and thus form a basis. Over $\mathbb{R}$, the vector space has dimension 2.

4. Dimension. Let K be a field, fix $g(X) := X + 5 \in K[X]$, and let $d \geq 1$. Compute the dimension of the space

$$W = \{h \in K[X] \mid \deg(h) \leq d \wedge \exists f \in K[X] : h = gf\}.$$

Lösung: Let

$$S = \{g, Xg, X^2g, \dots, X^{d-1}g\}.$$

These polynomials clearly belong to W since they are divisible by g and since

$$\deg(X^i g) = i + \deg(g) = i + 1,$$

which is smaller or equal to d for $0 \leq i \leq d-1$. Moreover, this is a linearly independent set as all these polynomials are of different degrees. Now, since

$$W \subset K[X]_{\leq d} := \{p \in K[X] \mid \deg(p) \leq d\},$$

we know that $\dim(W) \leq d+1$ (recall that $\{1, X, X^2, \dots, X^d\}$ is a basis of $K[X]_{\leq d}$ over K). Additionally, since there is no constant non-zero polynomial in W , its dimension is at most d . Since S is a set of d linearly independent polynomials in W , it is a basis of W and $\dim(W) = d$.

5. Dimension. Determine a basis and compute the dimension of the space

$$\{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is twice differentiable} \wedge (f + f'' = 0)\}.$$

Hint: Note that f, f' , and f'' exist and are continuous. Moreover, you can use the following result from Analysis: the function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $\forall x \in \mathbb{R} : f(x) = 0$ is the unique continuous solution to

$$\begin{cases} f + f'' &= 0 \\ f(0) &= f'(0) = 0 \end{cases}$$

Solution: We reformulate the hint: for every two times continuously differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(0) = 0$, $f'(0) = 0$ and $f(x) + f''(x) = 0 \forall x \in \mathbb{R}$, we have $f(x) = 0 \forall x \in \mathbb{R}$. Now let $V = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is } C^2 \text{ and } f + f'' = 0\}$ and let $f \in V$. Set $a := f(0)$ and $b := f'(0)$. Also define $g(x) := f(x) - a \cos(x) - b \sin(x)$. We have $g(0) = 0$, $g'(0) = 0$ and $g \in V$ as $\sin(x), \cos(x) \in V$. Thus, we have $g(x) = 0 \forall x \in \mathbb{R}$. Therefore, we find $f(x) = a \cos(x) + b \sin(x)$. Thus every element of V is a linear combination of $\sin(x)$ and $\cos(x)$. These functions are linearly independent elements over $\mathbb{R}$ and thus form a basis of V . We conclude that V has dimension 2.